

PRAVIN MANI KANNAN

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EDUCATION

Master of Science in Physics

Friedrich-Alexander-Universität Erlangen-Nürnberg

Apr 2024 - Present Erlangen, Germany

Master of Science in Nuclear Physics

University of Madras - CGPA: 8.96

Jul 2021 - Aug 2022 Chennai, India

Bachelor of Science in Physics

University of Madras - CGPA: 8.96

Jun 2018 - Apr 2021 Chennai, India

PHYSICS PROJECTS

Ising Model with Cython *[Independent Project]*

- Implemented a 2D Ising Monte Carlo with C-typed memory views and white-black Metropolis updates in Cython.
- Periodic boundary conditions and vector-free inner loops for substantial speedup over pure Python.

In Progress – Ising Phase Diagram with Reliable Uncertainty using JAX, FLAX *[Independent Project]*

- Building a FLAX-based CNN that applies conformal prediction to produce calibrated phase maps with confidence maps.

Comparative Analysis of Adiabatic Error Bounds in Quantum Systems Through a Modified Landau-Zener Model *[Graded: 1.5]*

FAU Erlangen-Nürnberg, under Prof. Michael J. Hartmann, Leonhard Richter, and Marco Wiedmann

- Constructed a three-level quantum system and observed the simulated errors from continuous and adiabatic evolution through their eigenlevels.
- Used a quantitative error upper bound to verify the simulated errors and test the bound itself from ‘Jansen, Ruskai, and Seiler’.

In Progress – Extending the Above Project to Discrete Evolution through Unitary Operators and Characterizing the Adiabatic Error Behavior *[Independent Project]*

- Constructed the unitary operator evolution through QuTiP.
- Currently testing out on the error bounds.

EXPERIMENTS

Determination of back scattering and absorption factor of Gamma and Beta rays using GM Counter

Gamma Ray Spectrometer: Characteristics

University of Madras, under Prof. Dr. A. Stephen

Fabry-Perot Interferometer: Multiple Beam Interference

Fiber Optics – Determination of Numerical Aperture, Estimation of Loss, Audio transmission

University of Madras, under Assistant Prof. Dr. J. Senthil Selvan

SKILLS

Language: German (A2), English (C1), Tamil (Native), Spanish (A1)

Programming Languages: Python, C++ (Beginner), Lisp

Machine Learning Libraries: TensorFlow, JAX, FLAX, OPTAX

Quantum System Libraries: QuTiP, Qiskit (Beginner)

Developer Tools / Framework: PyCharm, Emacs, Git, Jupyter, L^AT_EX